MAT1252
Mathematics for Computing

Lecture 05

Predicates & Sets

Outline

01. Predicates and Propositions
02. Quantifiers
03. Statements Negation
04. Propositions with two quantifiers
05. Negation of a proposition with two quantifiers
06. An algebraic derivation
07. Set Theory
08. The Addition Principle
09. Laws of Sets
10. Illustration of laws using Venn diagrams
11. Analogy with logic concepts
12. Cartesian Product

Lecture's Major Objectives

After completing this section, students should be able to:

- translate from a verbal statement involving **universal and existential quantifiers** into a symbolic statement, and vice versa
- find the negations of quantified statements
- draw **Venn diagrams** to represent **set expressions**
- find the complement of a set
- find the union and intersection of two sets
- find the Cartesian product of two sets and draw its graph
- find the power set of a set
- illustrate the equality of sets using Venn diagrams
- apply the Addition Principle

Predicates and Propositions

A predicate is a statement with one or more **variables**.

If we assign **values to the variables** then the statement becomes a **proposition**, and has a truth value.

Example

"$x > 20$" is a predicate, and if $x = 7$ then "$7 > 20$" becomes a proposition.

"x owns y cats" is a predicate, and if x = Peter and y = 3, then "Peter owns 3 cats" becomes a proposition.

More examples

If we define $P(x): x + 5 = 12$, then $P(3)$ is false and $P(7)$ is true.

If we define Q(x, y): the name x has y letters, then Q(John, 4) is true and Q(Sam, 8) is false.

Sets and set notation

A set is a collection of things, usually (but not necessarily) sharing some common attribute.

E.g. R = set of all people in this room = {Julia, David, Emily, Ali,…}
E.g. A = set of all letters in the alphabet = {a, b, c, d, …}
E.g. S = set of sheep that are also US presidents = $\emptyset$ ← empty set

We use the symbol $\in$ to denote membership of a set.
The symbol $\in$ is read "is a member of" or "is an element of" or "belongs to"

g $\in$ A
Julia Collins $\in$ R

% $\notin$ A
Justin Brown $\notin$ R

Special sets of numbers

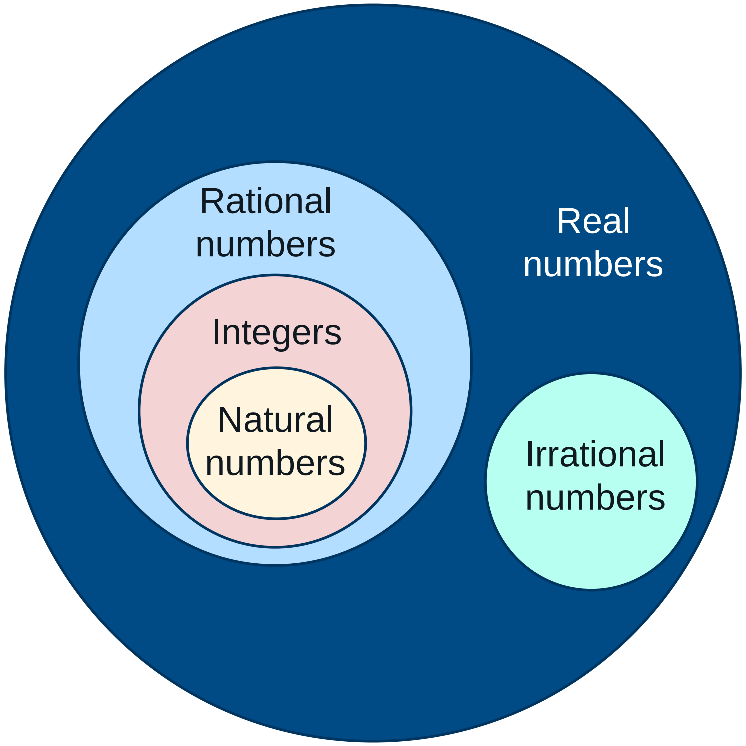
In maths we have some special sets of numbers.

$\mathbb{R}$ denotes the set of all real numbers (numbers with a decimal point, e.g. $\pi, \sqrt{2}$, 0.12345).

$\mathbb{Q}$ denotes the set of all rational numbers (numbers that can be written as a ratio of whole numbers p/q, i.e. fractions like 2/3)

$\mathbb{Z}$ denotes the set of all integers, i.e. positive and negative whole numbers, including zero = {…, -3, -2, -1, 0, 1, 2, 3, 4, …}.

$\mathbb{N}$ denotes the set of all natural numbers, i.e. positive whole numbers = {1, 2, 3, 4, …}.



Quantifiers

Instead of substituting particular values for the variables in predicates, we can make more general statements by using **quantifiers**.

We will use two quantifiers:

- Universal quantifier $\forall$**: "for all", "for every"**
- Existential quantifier $\exists$**: "there exists", "there is at least one"**

Quantifiers are used to link truth values of **predicates** to **sets**.

A typical use of a quantifier is:

- For every element of a set S, the predicate P is true: $\forall\ s \in S, P(s)$
- There is an element of a set S for which the predicate P is true:
$$\exists\ s \in S, P(s)$$

Universal quantifiers

The universal quantifier $\forall$ is used for statements of the type

"All do … " or "None do … "

"All are … " or "None are … “

Example 1a

If R = set of people in this room and the predicate C is C(x): x likes chocolate then: $\forall x \in R, C(x)$

means "Everybody in this room likes chocolate".

Example 2a

If the predicate D is D(x): x likes going to the dentist then:

$$\forall x \in P, \sim D(x)$$

means "Everybody here dislikes going to the dentist", or "Nobody in this room likes going to the dentist".

Existential quantifiers

The existential quantifier $\exists$ is used for statements of the type

"Some do … " or "Some don't … "

"Some are … " or "Some are not … "

Example 1b

If R = set of people in this room and we define the predicate I as I(x): x speaks Italian then: $\exists\, \mathbf{x} \in \mathbf{R}, \mathbf{I(x)}$

means "There is at least one person in this room who speaks Italian".

Example 2b

The statement $\exists\, \mathbf{x} \in \mathbf{R}, \mathbf{\sim I(x)}$

means "There is at least one person in this room who does not speak Italian".

More examples of quantifiers

Example 3

If we define the predicate $S(x): (x > 2) \wedge (x < 7)$, then:

$$\exists\, x \in \mathbb{N},\ S(x)$$

means "There is at least one natural number that is greater than 2 and less than 7".

Example 4

If we define the predicate $Q(x): (x < 5) \vee (x \geq 5)$, then:

$$\forall\, x \in \mathbb{R},\ Q(x)$$

means "Every real number is either less than 5 or is greater than or equal to 5".

Negation of quantifiers ($\forall$)

The negation of a statement is another statement that contradicts it.

Example

Consider the statement "All swans are black." We would contradict this by finding an example of a swan that was not black.

In symbols: let **S** = set of swans, and B be the predicate **B(x)**: Swan x is black.

The statement "All swans are black: is $\forall x \in S, B(x)$

Its negation is "There exists a swan that is not black", i.e. $\exists x \in S, \sim B(x)$

So we have $\sim(\forall x \in S, B(x)) \equiv \exists x \in S, \sim B(x)$

Notice that the **negation of** $\forall$ **gives** $\exists$.

Or, put another way:

- the opposite of "All do" is "Some do not"
- the opposite of "All are" is "Some are not"

Negation of $\forall$

Suppose R = Set of people in this room and define:

$P(x)$: person x is asleep

Then the claim is: $\forall x \in R, P(x)$ (everyone in the room is asleep)

and it is refuted by: $\exists x \in R, \sim P(x)$ (there is someone in the room who is not asleep)

So the negation of claim

$$\forall x \in R, P(x) \text{ is } \exists x \in R, \sim P(x).$$

That is

$$\sim(\forall x \in R, P(x)) \equiv \exists x \in R, \sim P(x)$$

Negation of quantifiers (∃)

The negation of a statement is another statement that contradicts it.

Example

Consider the statement "There is a swan which is black." We would contradict this by showing that all of the swans are not black, i.e. that none of the swans are black.

In symbols: let **S** = set of swans, and B be the predicate **B(x)**: Swan x is black.

The statement "There is a swan which is black: is $\exists\, x \in S, B(x)$

Its negation is "All swans are not black", i.e. $\forall\, x \in S, \sim B(x)$

So we have $\sim(\exists\, x \in S, B(x)) \equiv \forall\, x \in S, \sim B(x)$

Notice that the **negation of** ∃ **gives** ∀.

Or, put another way:

- The opposite of "Some do" is "None do"
- The opposite of "Some are" is "None are"

Negation of ∃

Suppose R = Set of people in this room and define:

P(x): person x is asleep

Then the claim is: $\exists\, \mathbf{x} \in \mathbf{R}, \mathbf{P(x)}$ (someone in this room is asleep)

and it is refuted by: $\forall\, \mathbf{x} \in \mathbf{R}, \mathbf{\sim P(x)}$ (everyone in this room is awake)

So the negation of claim

$\exists\, \mathbf{x} \in \mathbf{R}, \mathbf{P(x)}$ is $\forall\, \mathbf{x} \in \mathbf{R}, \mathbf{\sim P(x)}$.

That is

$$\mathbf{\sim(\exists\, x \in R, P(x))} \equiv \mathbf{\forall\, x \in R, \sim P(x)}$$

Summary of negating quantifiers

Statement	Negation
All do …	Some don't
All are …	Some are not …
Some do …	None do …
Some are …	None are …

Statements Negation

Negate each of the following statements:

- All cars run on petrol.
- Some files are not binary.
- Everybody likes ice cream.
- Some students like Maths.
- Nobody listens to Beethoven.
- Some people do not eat curry.
- All rabbits are grey.
- Not all dogs bite.

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  Everybody eats curry
- All rabbits are grey.
  Some rabbits are not grey.
- Not all dogs bite.
  All dogs bite.

Propositions with two quantifiers

Now we consider propositions with two quantifiers.

Example

Let P be a set of people and M a set of movies, and define the predicate S(x, y) to mean "person x has seen movie y".

Consider the following symbolic statements:

(i) $\forall x \in P, \exists y \in M, S(x, y)$

which may be translated:

"For each person x, there is some movie y such that x has seen y."

or in simpler English:

"Every person has seen at least one movie."

Propositions with two quantifiers

(ii) $\exists\, x \in P, \forall\, y \in M, S(x, y)$

which may be translated:

or more simply:

(iii) $\exists\, x \in P, \exists\, y \in M, S(x, y)$

which may be translated:

or in simpler English:

(iv) $\forall\, x \in P, \forall\, y \in M, S(x, y)$

which may be translated:

or more simply:

Propositions with two quantifiers

(ii) $\exists\, x \in P, \forall\, y \in M, S(x, y)$

which may be translated:

"There is some person x such that for every movie y, x has seen y."

or more simply:

"Some person has seen every movie."

(iii) $\exists\, x \in P, \exists\, y \in M, S(x, y)$

which may be translated:

or in simpler English:

(iv) $\forall\, x \in P, \forall\, y \in M, S(x, y)$

which may be translated:

or more simply:

Propositions with two quantifiers

(ii) $\exists x \in P, \forall y \in M, S(x, y)$

which may be translated:

"There is some person x such that for every movie y, x has seen y."

or more simply:

"Some person has seen every movie."

(iii) $\exists x \in P, \exists y \in M, S(x, y)$

which may be translated:

"There is some person x such that for some movie y, x has seen y."

or in simpler English:

"Some person has seen some movie."

(iv) $\forall x \in P, \forall y \in M, S(x, y)$

which may be translated:

or more simply:

Propositions with two quantifiers

(ii) $\exists\, x \in P, \forall\, y \in M, S(x, y)$

which may be translated:

"There is some person x such that for every movie y, x has seen y."

or more simply:

"Some person has seen every movie."

(iii) $\exists\, x \in P, \exists\, y \in M, S(x, y)$

which may be translated:

"There is some person x such that for some movie y, x has seen y."

or in simpler English:

"Some person has seen some movie."

(iv) $\forall\, x \in P, \forall\, y \in M, S(x, y)$

which may be translated:

"For every person x and for every movie y, x has seen y."

or more simply:

"Every person has seen every movie."

Propositions with two quantifiers

Notice that if we simply replace $\forall$ by "for all" and $\exists$ by "there exists", the resulting statement may be very awkward, and we may need to insert the phrase "such that" after $\exists$ in order to obtain a reasonable sentence.

Notice also that the order of the quantifiers is usually significant.

Using the above example again, **"$\exists$ y $\in$ M, $\forall$ x $\in$ P, S(x, y)"**

means there is at least one movie that every person has seen,

whereas **"$\forall$ x $\in$ P, $\exists$ y $\in$ M, S(x, y)"**

means each person has seen at least one movie.

We can see the difference if we assume that the diagram below shows which people have seen various movies.

Propositions with two quantifiers

For the situation shown on the diagram, the statement

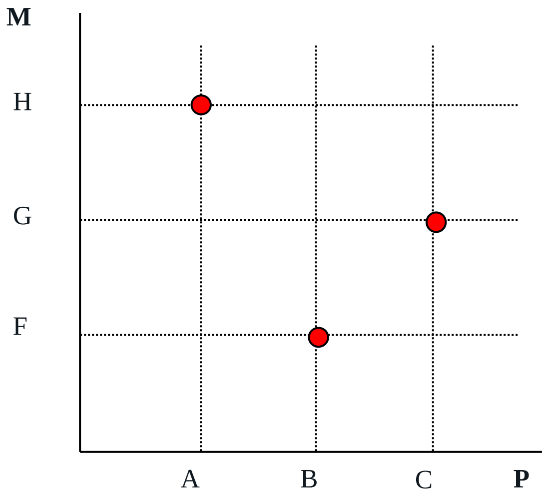
$$\exists\, y \in M, \forall\, x \in P, S(x, y)$$

is false, because there is no movie that everyone has seen,

whereas the statement

$$\forall\, x \in P, \exists\, y \in M, S(x, y)$$

is true, since every person has seen at least one movie.



Negation of a proposition with two quantifiers

Example

Let $P = \{A, B, C\}$ be a set of people

and $M = \{F, G, H\}$ be a set of movies,

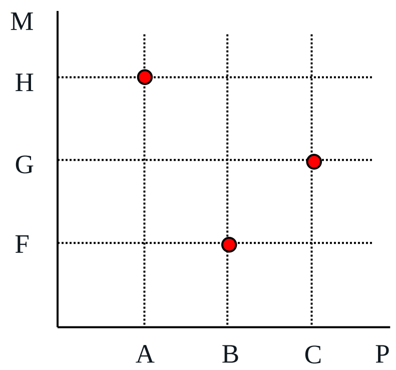
and consider the predicate

$S(x, y)$ means "person x has seen movie y".

To negate a proposition, change every ∀ to ∃ and vice versa, and negate the final predicate.

Negation of a proposition with two quantifiers

(i) Negation of $\forall$ $\exists$



Consider the claim:

$$\forall\, x \in P, \exists\, y \in M, S(x, y)$$

In simple English this is:

"Every person has seen at least one movie."

The negation of that claim is: $\exists\, x \in P, \forall\, y \in M, \sim S(x, y)$.

There exists some person such that, for each movie, that person has not seen the movie

or there is some person who has not seen any movie.

$$\sim(\forall\, x \in P, \exists\, y \in M, S(x, y)) \equiv \exists\, x \in P, \forall\, y \in M, \sim S(x, y)$$

Negation of a proposition with two quantifiers

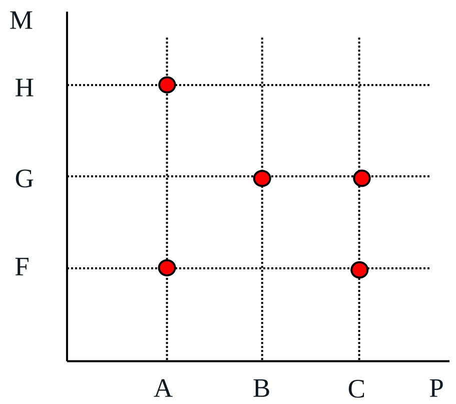
(ii) Negation of $\exists\ \forall$

Consider the claim:

$$\exists\ y \in M,\ \forall\ x \in P,\ S(x, y)$$

In simple English this is:

"There is some movie that everyone has seen."



The negation of that claim is: $\forall\ y \in M,\ \exists\ x \in P,\ {\sim}S(x, y)$.

For each movie there is someone who has not seen it

or no movie has been seen by everybody.

$$\sim(\exists\ y \in M,\ \forall\ x \in P,\ S(x, y)) \equiv \forall\ y \in M,\ \exists\ x \in P,\ {\sim}S(x, y)$$

Negation of a proposition with two quantifiers

(i) Negation of $\forall \, \exists$

$$\sim(\forall x \in P, \exists y \in M, S(x, y)) \equiv \exists x \in P, \forall y \in M, \sim S(x, y)$$

(ii) Negation of $\exists \, \forall$

$$\sim(\exists y \in M, \forall x \in P, S(x, y)) \equiv \forall y \in M, \exists x \in P, \sim S(x, y)$$

(iii) Negation of $\exists \, \exists$

$$\sim(\exists x \in P, \exists y \in M, S(x, y)) \equiv \forall x \in P, \forall y \in M, \sim S(x, y)$$

(vi) Negation of $\forall \, \forall$

$$\sim(\forall y \in M, \forall x \in P, S(x, y)) \equiv \exists y \in M, \exists x \in P, \sim S(x, y)$$

An algebraic derivation

The laws for the negation of doubly-quantified propositions may be derived algebraically.

Examples

Find the negation of each of the following propositions by:

(i) translating the proposition into symbols,

(ii) finding the negation of the symbolic statement

(iii) translating the negated symbolic statement into simple English.

1. For each movie there is some person who has seen it.
2. Somebody has seen every movie.
3. There is some movie that everyone has seen.

An algebraic derivation

1. For each movie there is some person who has seen it.

(i)

(ii)

(iii)

2. Somebody has seen every movie.

(i)

(ii)

(iii)

3. There is some movie that everyone has seen.

(i)

(ii)

(iii)

An algebraic derivation

1. For each movie there is some person who has seen it.

(i) $\forall\, y \in M, \exists\, x \in P,\ S(x, y)$

(ii) $\exists\, y \in M, \forall\, x \in P, \sim S(x, y)$

(iii) There is at least one movie that nobody has seen.

2. Somebody has seen every movie.

(i)

(ii)

(iii)

3. There is some movie that everyone has seen.

(i)

(ii)

(iii)

An algebraic derivation

1. For each movie there is some person who has seen it.

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2. Somebody has seen every movie.

(i) $\exists x \in P, \forall y \in M, S(x, y)$

(ii) $\forall x \in P, \exists y \in M, \sim S(x, y)$

(iii) Nobody has seen every movie.

3. There is some movie that everyone has seen.

(i)

(ii)

(iii)

An algebraic derivation

1. For each movie there is some person who has seen it.

(i) $\forall\, \mathbf{y} \in \mathbf{M}, \exists\, \mathbf{x} \in \mathbf{P},\ \mathbf{S(x, y)}$

(ii) $\exists\, \mathbf{y} \in \mathbf{M}, \forall\, \mathbf{x} \in \mathbf{P}, \sim\mathbf{S(x, y)}$

(iii) There is at least one movie that nobody has seen.

2. Somebody has seen every movie.

(i) $\exists\, \mathbf{x} \in \mathbf{P}, \forall\, \mathbf{y} \in \mathbf{M},\ \mathbf{S(x, y)}$

(ii) $\forall\, \mathbf{x} \in \mathbf{P}, \exists\, \mathbf{y} \in \mathbf{M}, \sim\mathbf{S(x, y)}$

(iii) Nobody has seen every movie.

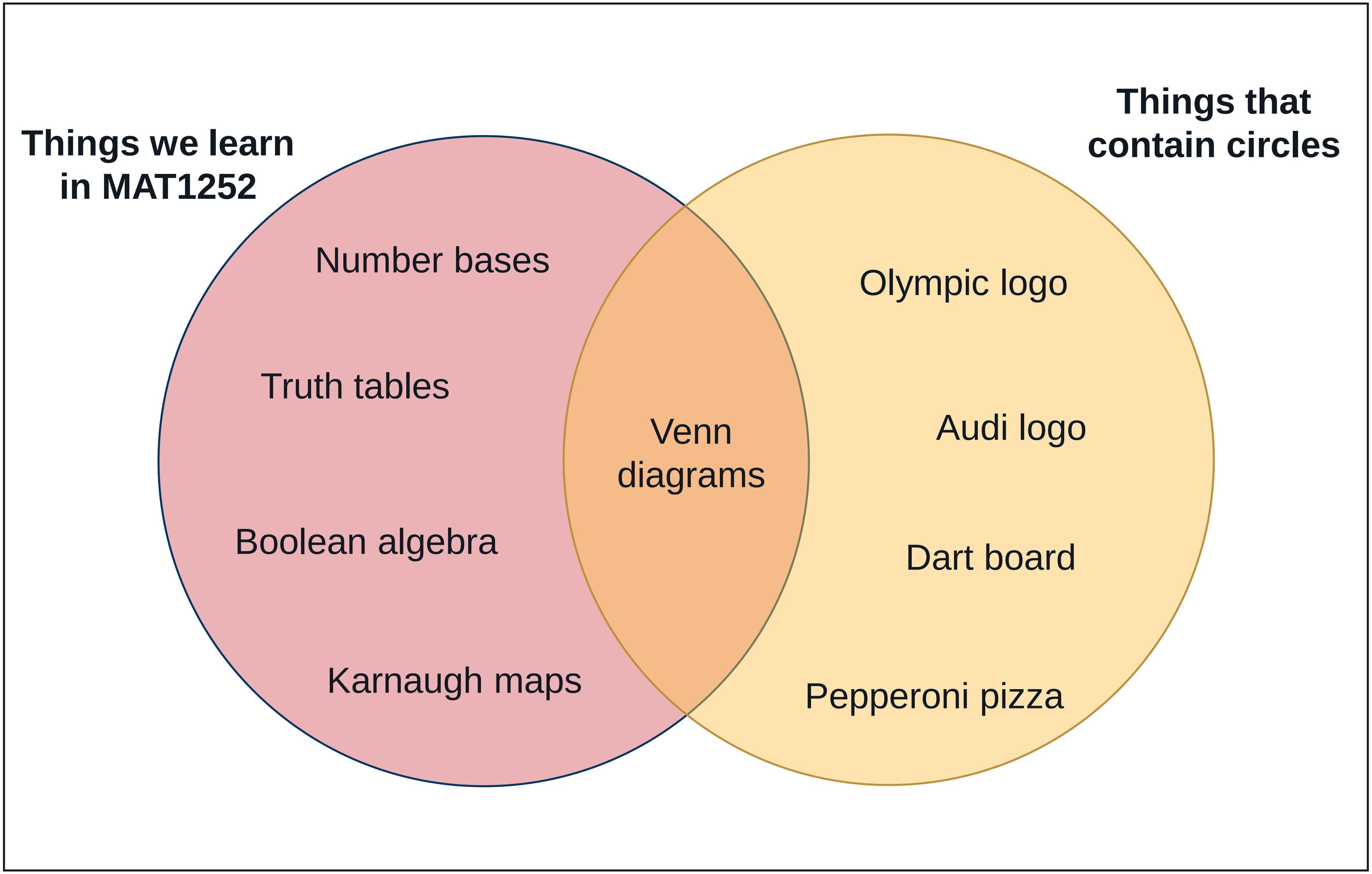
3. There is some movie that everyone has seen.

(i) $\exists\, \mathbf{y} \in \mathbf{M}, \forall\, \mathbf{x} \in \mathbf{P},\ \mathbf{S(x, y)}$

(ii) $\forall\, \mathbf{y} \in \mathbf{M}, \exists\, \mathbf{x} \in \mathbf{P}, \sim\mathbf{S(x, y)}$

(iii) For each movie there is somebody who has not seen it.


ALTERNATIVE HERITAGE
Mathematician,
Philosopher &
Anglican priest
JOHN
VENN
ALTERNATIVE HERITAGE
Really strong
beard game


Things we learn in MAT1252
Things that contain circles
Number bases
Truth tables
Boolean algebra
Karnaugh maps
Venn diagrams
Olympic logo
Audi logo
Dart board
Pepperoni pizza

Set Theory - Sets

A set is a collection of things, usually (but not necessarily) sharing some common attribute.

For example, you might be concerned with a set of files, a set of dates, a set of orders for a particular month, a set of customers, and so on.

P = {Word, Excel, Powerpoint, Outlook}
E = {2, 4, 6, 8, 10, …}
S = {x | x is an even number < 18}
D = {d | d is a date before 1 July}

Recall that in maths we have some special sets of numbers

$\mathbb{N}$ denotes the set of all natural numbers = {1, 2, 3, 4, …}
$\mathbb{Z}$ denotes the set of all integers = {0, ±1, ±2, ±3, ±4,…}
$\mathbb{R}$ denotes the set of all real numbers (numbers with a decimal point)

Specification of a set

There are various ways to specify a set.

For example, we can list the members of a set:

$\{1, 2, 3, 4\}$

$\{1, 2, 3, 4, \ldots\}$

$\{1, 2, 3, 4, \ldots, 30\}$

Or we can describe the members of the set

$\{x \mid x \text{ is an odd number less than } 10\}$

$\{y \in N \mid y > 5 \text{ and } y < 13\}$

The order of the elements of a set is unimportant, so that $\{a, c, b\}$ and $\{a, b, c\}$ are the same set.

Repetitions are unimportant, so the set $\{a, a, a, b, a, c, c, b\}$ is the same as $\{a, b, c\}$.

The Universal Set and the Empty Set

The universal set **U** is the set of **all the things** that we happen to be working with in a particular problem.

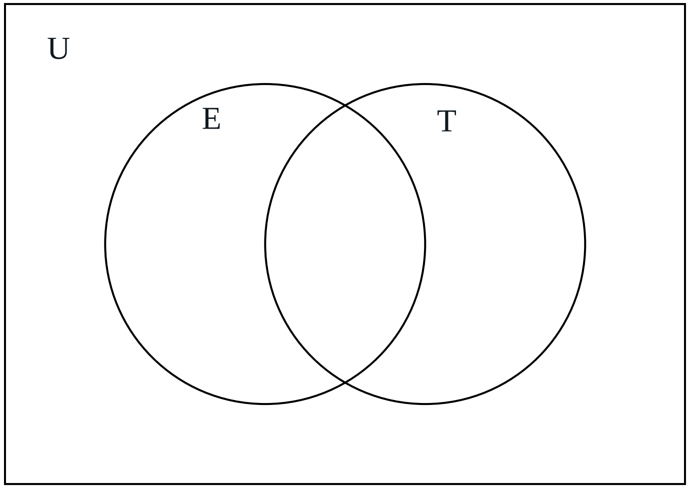
It is sometimes convenient to talk about the empty set $\varnothing$, or $\{ \}$, which is a set that has no members.

Membership of a set

The symbol $\in$ is read "is a member of" or "is an element of" or "belongs to"

p $\in$ {p, q, r}, 3 $\in \mathbb{N}$, 4.5 $\notin \mathbb{N}$, $\pi \in \mathbb{R}$, $\sqrt{-1} \notin \mathbb{R}$

Examples of sets and Venn diagrams



Example 1

U = {1, 2, 3, …, 12}

E = {even numbers}

T = {numbers divisible by 3}

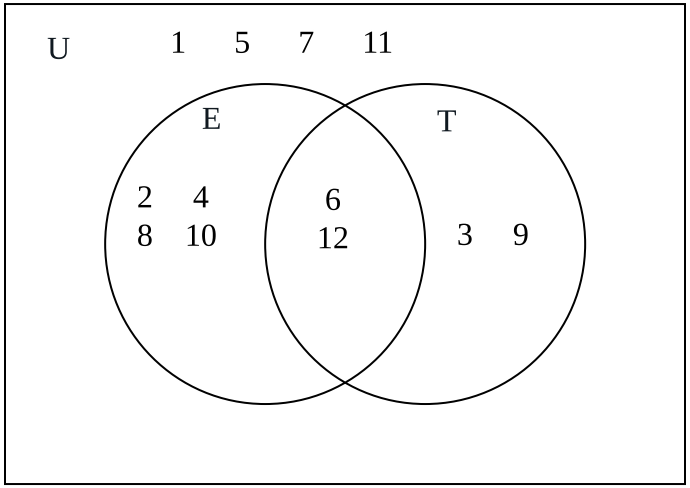
Place each number in the appropriate region.

Membership of a set

The symbol $\in$ is read "is a member of" or "is an element of" or "belongs to"

p $\in$ {p, q, r}, 3 $\in \mathbb{N}$, 4.5 $\notin \mathbb{N}$, $\pi \in \mathbb{R}$, $\sqrt{-1} \notin \mathbb{R}$

Examples of sets and Venn diagrams



Example 1

U = {1, 2, 3, …, 12}

E = {even numbers}

T = {numbers divisible by 3}

Place each number in the appropriate region.

Membership of a set

Example 2:

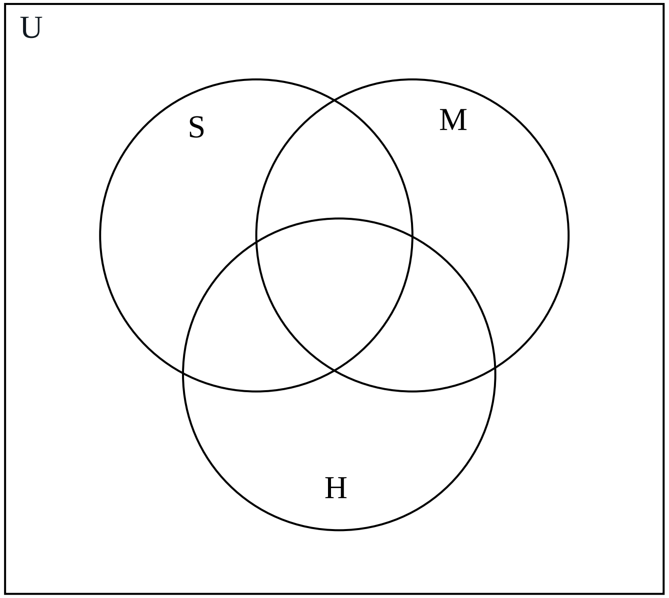
U = {animals}

H = {animals with horns}

M = {meat-eaters}

S = {striped animals}

Place in the appropriate subset: cow, horse, lion, tiger, zebra



Membership of a set

Example 2:

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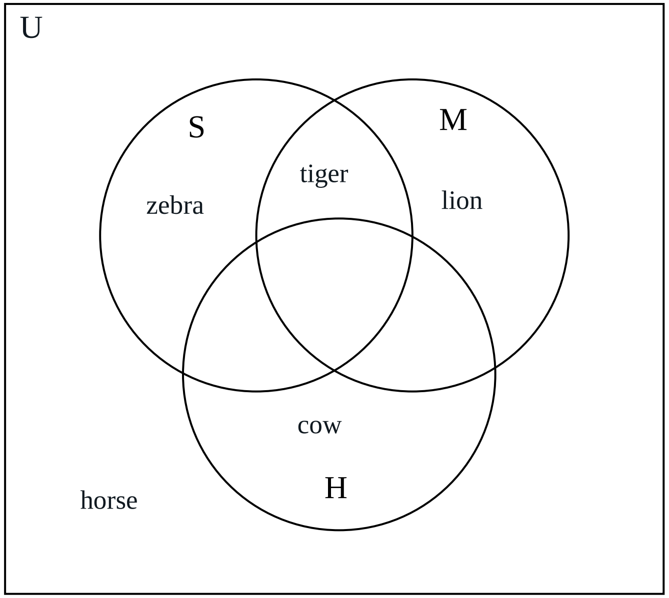
S = {striped animals}

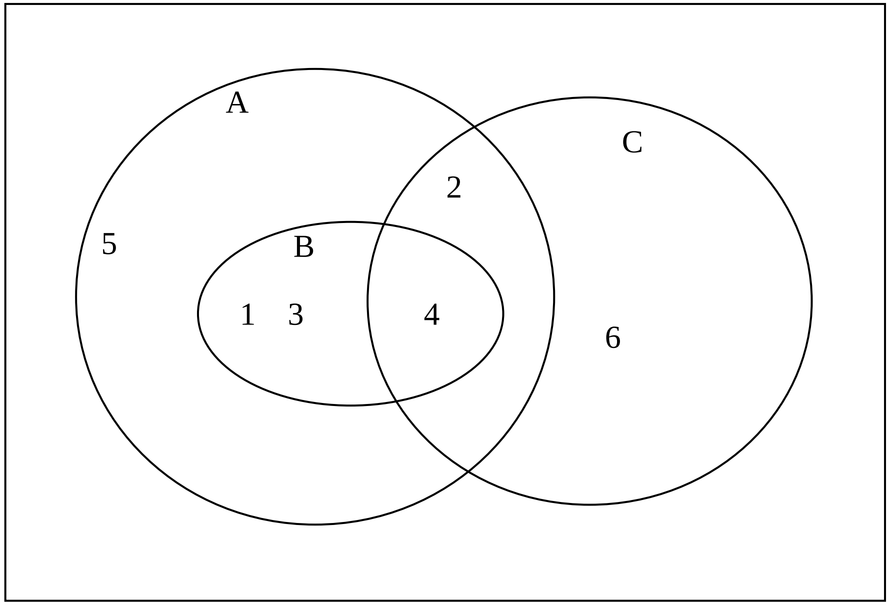
Place in the appropriate subset: cow, horse, lion, tiger, zebra



Subsets

We say that B is a subset of A, if every element of B is also an element of A. In symbols $B \subseteq A$.

Example: Let $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 3, 4\}$, $C = \{2, 4, 6\}$

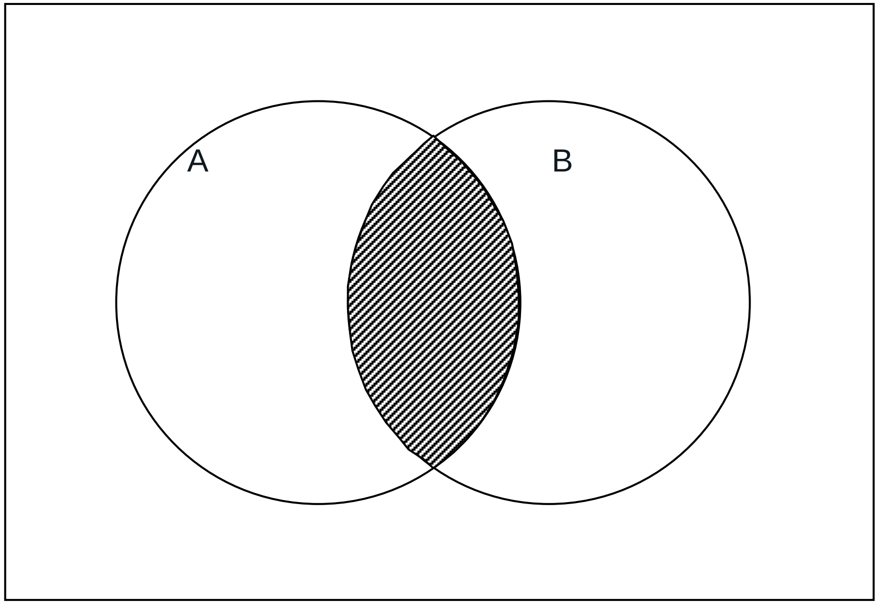


Clearly $B \subseteq A$, since each element of B is also an element of A.
Also clearly C is not a subset of A, since $6 \in C$ but $6 \notin A$.

Set operations: Intersection

$A \cap B$ is the set of elements in both A and B.

That is $\quad A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$

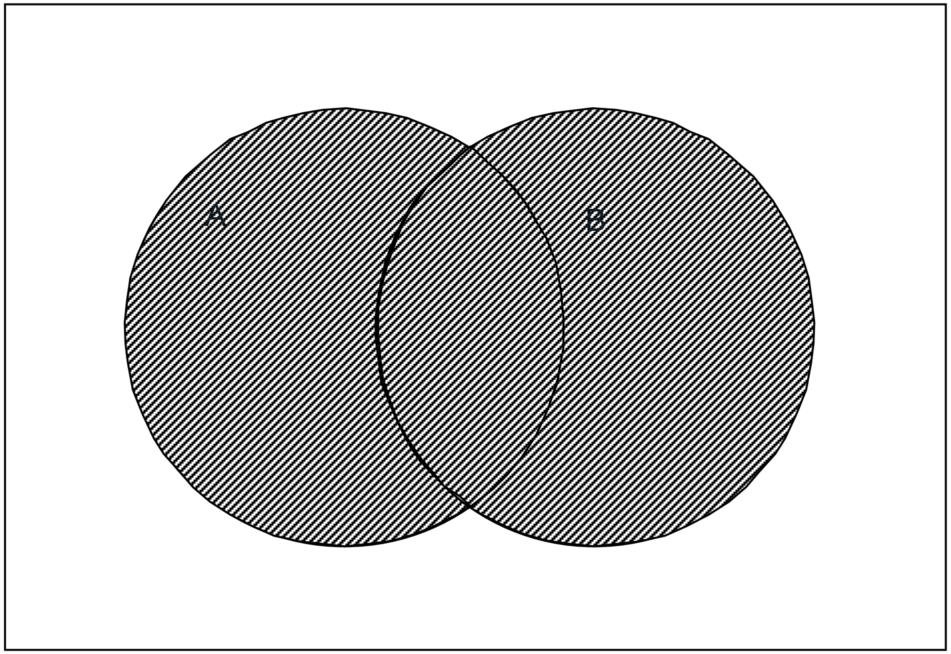


If A and B have no elements in common, that is if $A \cap B = \varnothing$, then A and B are said to be **disjoint**.

Set operations: Union

$A \cup B$ is the set of elements in A or B, or both.

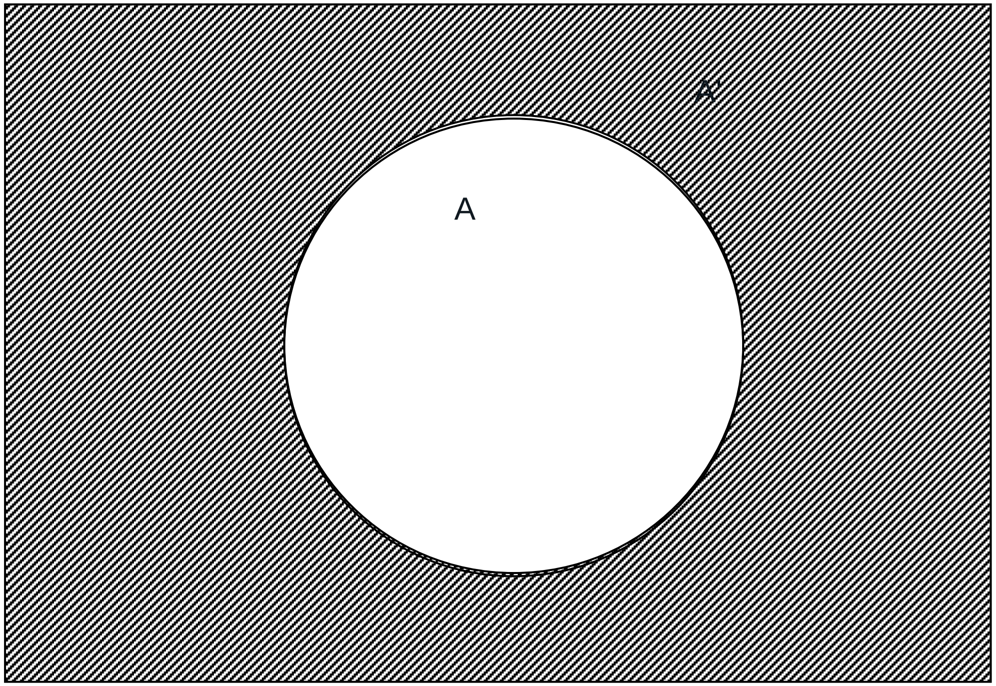
That is $\quad A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$



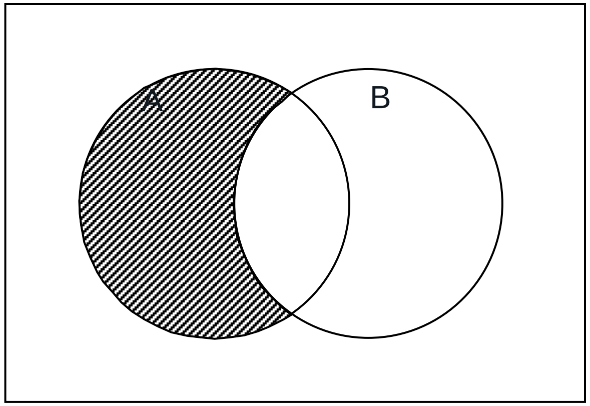
Set operations: Complement

The complement of A is denoted by A' and consists of all the elements of the universal set that are not in A.

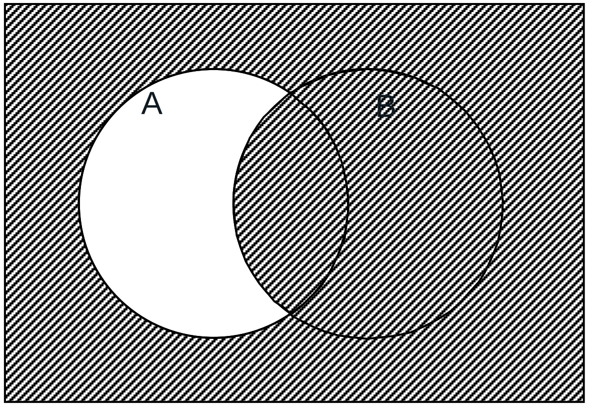
That is $\mathbf{A' = \{x \mid x \in U \wedge x \notin A\}}$



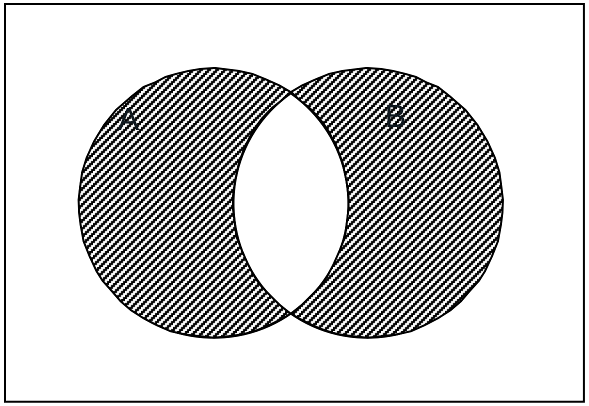
Set operations - example



$A \cap B'$

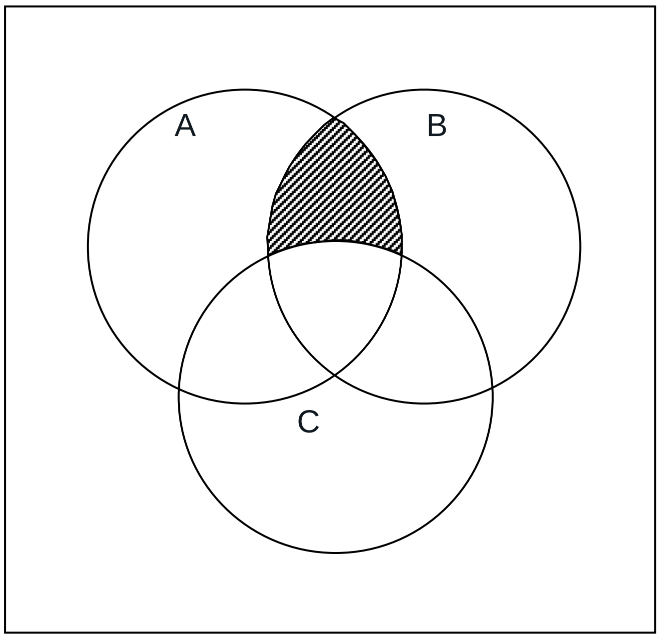


$A' \cup B \equiv (A \cap B')'$

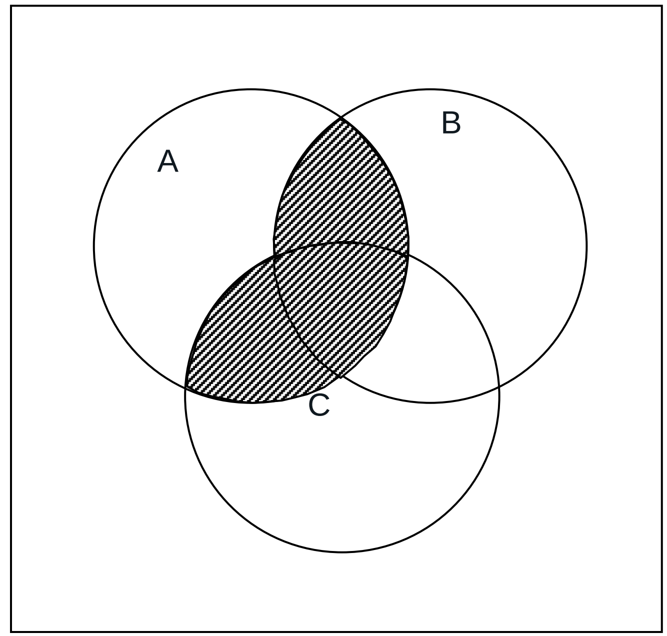


$(A \cap B') \cup (A' \cap B)$

Set operations - example

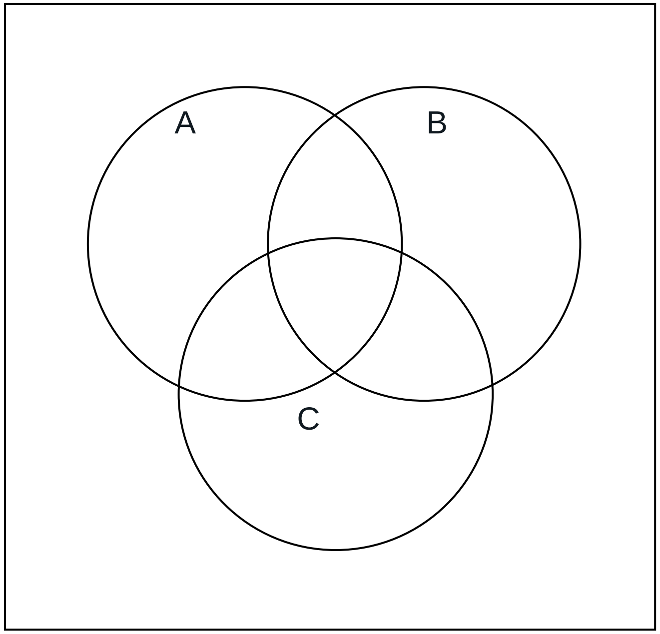


$A \cap B \cap C'$

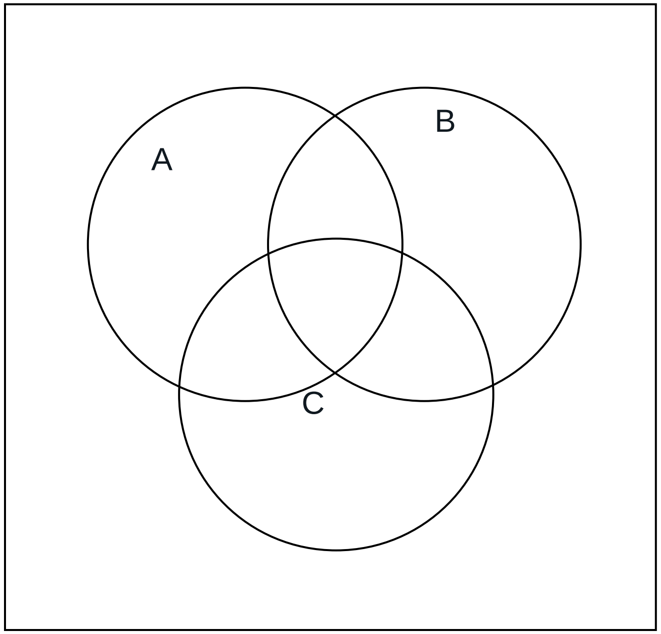


$A \cap (B \cup C)$

Set operations - example

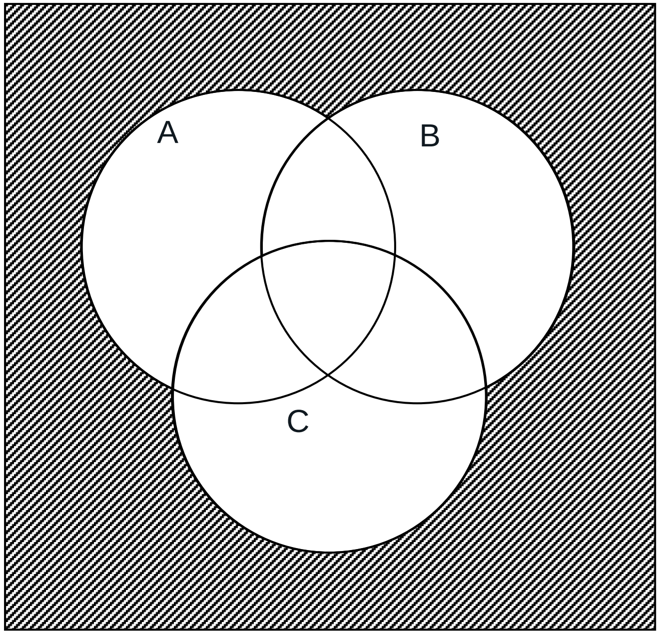


$A' \cap B' \cap C'$

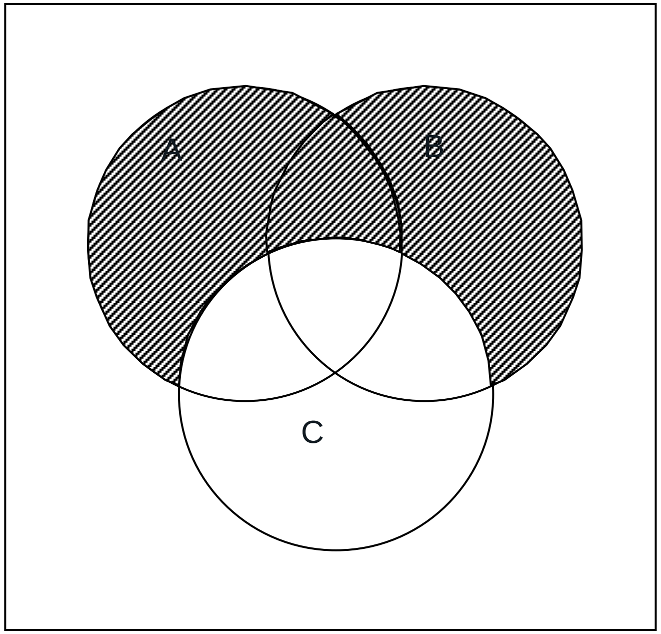


$(A \cup B) \cap C'$

Set operations - example

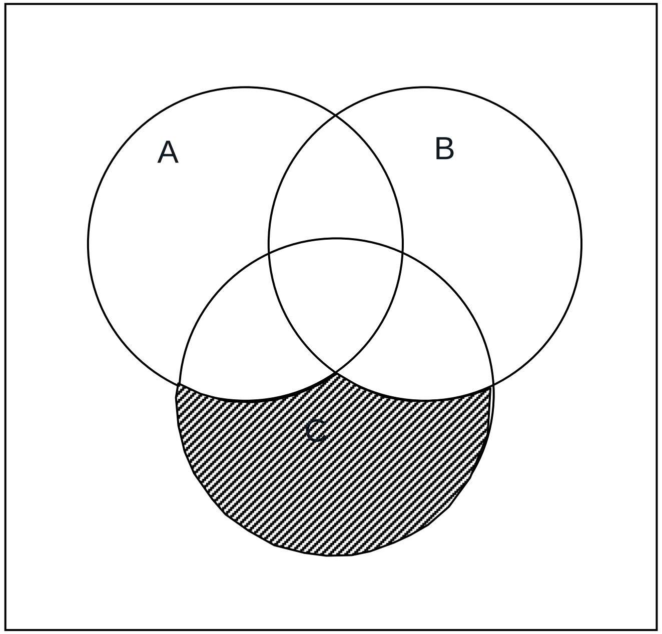


$A' \cap B' \cap C'$

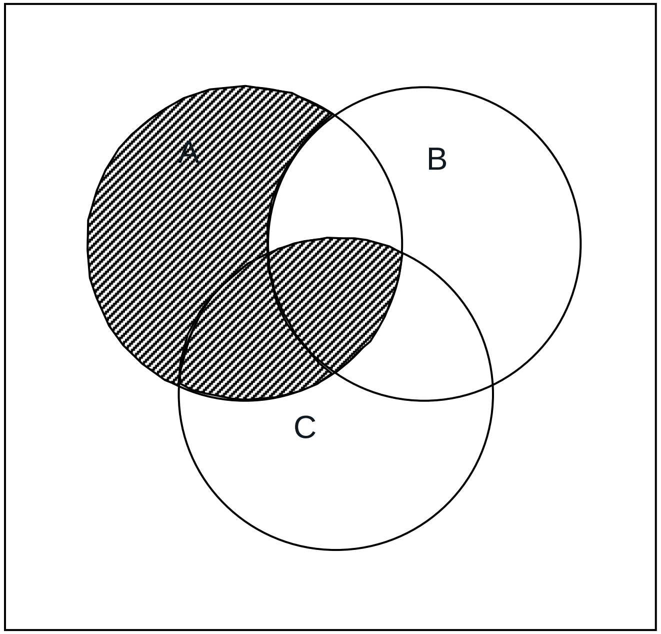


$(A \cup B) \cap C'$

Set operations - example



$C \cap (A \cup B)' \equiv C \cap A' \cap B'$



$(A \cap B') \cup (A \cap C)$

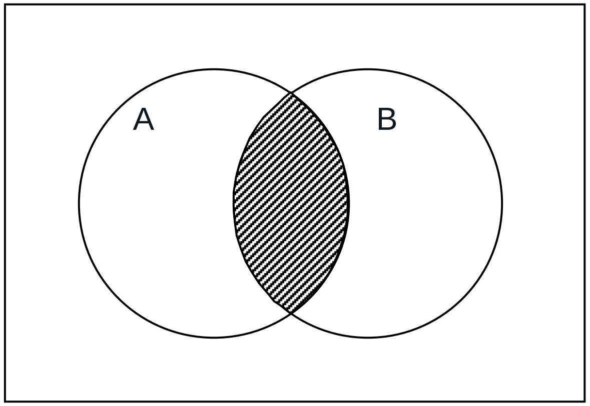
The Addition Principle

Let $n(A)$ denote the number of elements in set A.
If set A has $n(A)$ elements and set B has $n(B)$ elements, then the number of elements in the union $A \cup B$ is

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

This is called The Addition Principle.

From a Venn diagram we can see that if we add $n(A)$ and $n(B)$ then we will be counting $n(A \cap B)$ twice.



The Addition Principle

If A and B are **disjoint**, that is, if $A \cap B = \varnothing$ and hence $n(A \cap B) = 0$, then The Addition Principle becomes:

$$\mathbf{n(A \cup B) = n(A) + n(B)}$$

Example 1

Suppose that

A = {multiples of 3 less than 32}

A = {3, 6, 9, 12, 15, 18, 21, 24, 27, 30},

and

B = {multiples of 5 less than 32}

B = {5, 10, 15, 20, 25, 30}

Then

$A \cup B = \{3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30\}$

By counting: $n(A \cup B) = 14$

The Addition Principle - Examples

Example 2

$A = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30\}$
$B = \{5, 10, 15, 20, 25, 30\}$
$A \cap B = \{15, 30\}$

$n(A) = 10$, $n(B) = 6$, $n(A \cap B) = 2$

Using The Addition Principle:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$= 10 + 6 - 2 = 14$$

The Addition Principle - Examples

Example 3

$A = \{\text{multiples of 7 less than 50}\}$
$A = \{7, 14, 21, 28, 35, 42, 49\}$,
and
$B = \{\text{multiples of 12 less than 50}\}$
$B = \{12, 24, 36, 48\}$

Then $A \cup B = \{7, 12, 14, 21, 24, 28, 35, 36, 42, 48, 49\}$

By counting: $n(A \cup B) = 11$

Alternatively, using the addition principle,
$n(A) = 7$, $n(B) = 4$, $n(A \cap B) = 0$ (since $A \cap B = \varnothing$),

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$= 7 + 4 - 0 = 11$$

The Addition Principle - Examples

Example 4

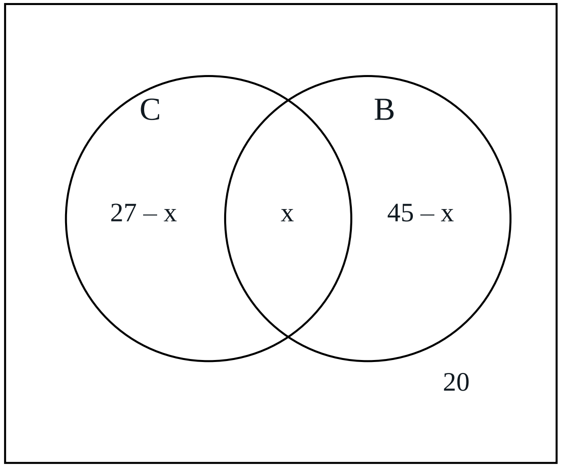
In a group of 80 students, 27 play cricket, 45 play baseball, and 20 play neither cricket nor baseball.

Find how many play:

(i) both cricket and baseball;
(ii) either cricket or baseball;
(iii) baseball, but not cricket;
(iv) cricket, but not baseball.

The Addition Principle - Examples

Solution:



Suppose that $n(C \cap B) = x$. Then,
since $n(C) = 27$, $n(C \cap B') $ must be $27 - x$,
and since $n(B) = 45$, $n(C' \cap B)$ must be $45 - x$.
Adding the expressions in the diagram, we get
$(27 - x) + x + (45 - x) + 20 = 80$
$92 - x = 80$
so $x = 12$, and from here it is easy to find the required numbers.

(i) both cricket and baseball: 12
(ii) either cricket or baseball: $(27 - 12) + 12 + (45 - 12) = 60$
(iii) baseball, but not cricket: $45 - 12 = 33$
(iv) cricket, but not baseball: $27 - 12 = 15$

The Addition Principle - Examples

Example 5

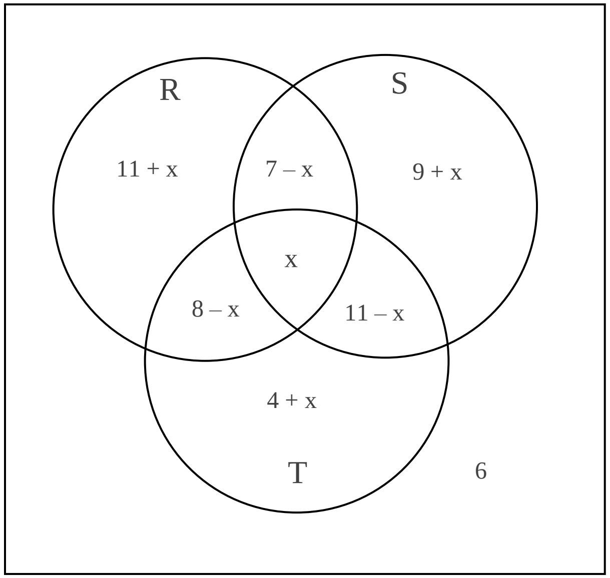
A poll of three TV shows, Rodeo, Starbus and Totem showed that in a group of 60 people, 26 watched Rodeo, 27 watched Starbus, 23 watched Totem, 7 watched Rodeo and Starbus, 8 watched Rodeo and Totem, 11 watched Starbus and Totem, and 6 watched none of the shows.

Find how many people watched:

(i) all three shows;
(ii) Rodeo and Starbus but not Totem;
(iii) Starbus, but not the other two shows;
(iv) only one of the three shows.

The Addition Principle - Examples

Solution:



We are given that:

$n(U) = 60$

$n(R) = 26$, $n(S) = 27$, $n(T) = 23$,

$n(R \cap S) = 7$, $n(R \cap T) = 8$, $n(S \cap T) = 11$

$n(R' \cap S' \cap T') = 6$

Suppose that x people watch all three.

Now refer to the diagram.

Since $n(R \cap S) = 7$, it follows that

$$n(R \cap S \cap T') = 7 - x$$

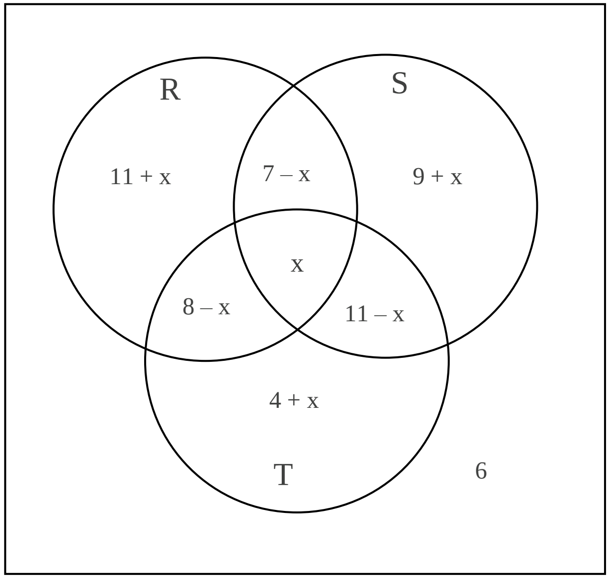
Similarly,

$$n(R \cap S' \cap T) = 8 - x$$

and

$$n((R' \cap S \cap T) = 11 - x$$

The Addition Principle - Examples



Again, referring to the diagram,

$n(R \cap S' \cap T') + (7 – x) + (8 – x) + x = 26$

and so

$n(R \cap S' \cap T') = 26 – (7 – x) – (8 – x) – x$

$= 11 + x$

Similarly $n(R' \cap S' \cap T) = 4 + x$

And $n(R' \cap S \cap T') = 9 + x$

Now, adding all the expressions in the diagram:

$(11 + x) + (7 – x) + (8 – x) + x + (4 + x) + (11 – x) + (9 + x) + 6 = 60$

which simplifies to $56 + x = 60$ and so $x = 4$

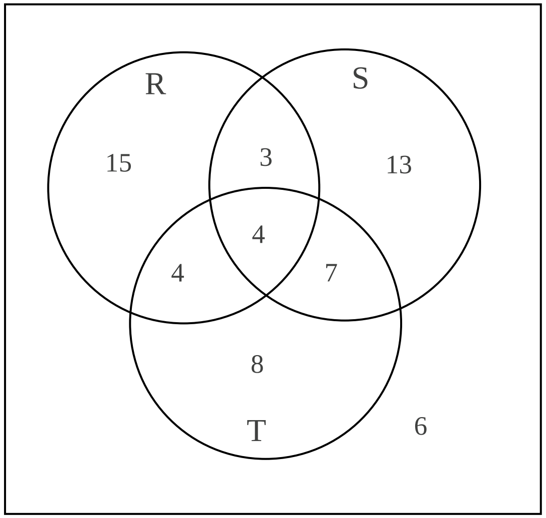
The answers are:

(i) 4 watch all three shows;

(ii) 3 watch Rodeo and Starbus but not Totem;

(iii) 13 watch Starbus only;

(iv) 36 watch only one of the three shows.



Laws of Sets

The most important are the distributive laws:

$$\mathbf{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}$$
$$\mathbf{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$$

and de Morgan's laws

$$\mathbf{(A \cup B)' = A' \cap B'}$$
$$\mathbf{(A \cap B)' = A' \cup B'}$$

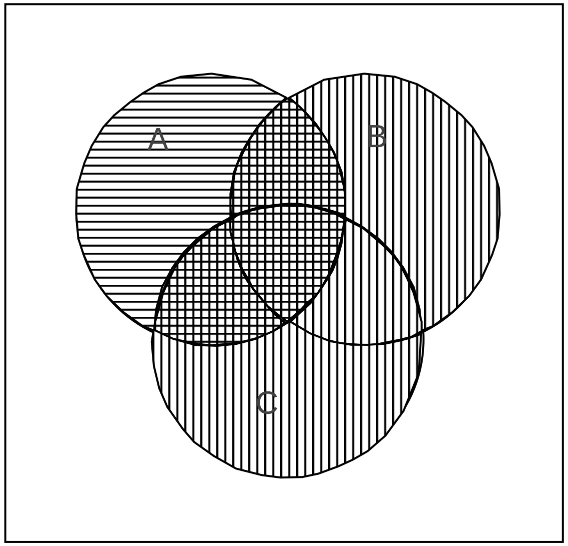
Notice that the laws are in pairs, and we can swap from a law to its dual (matching partner of the pair) by swapping $\cup$ with $\cap$, and U with $\varnothing$.

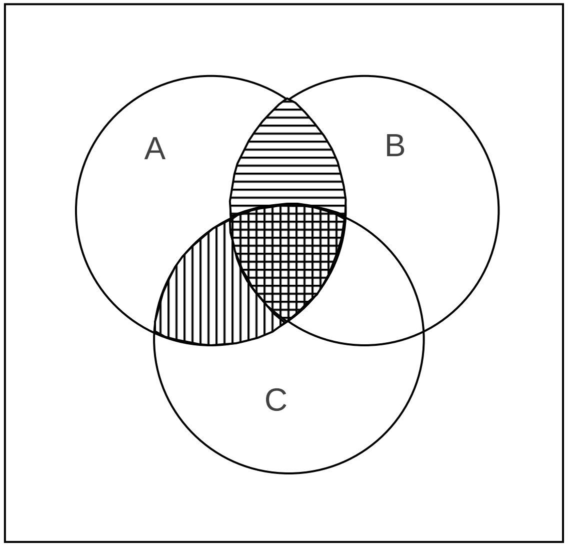
Illustration of laws using Venn diagrams

Consider, for example, the law

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

We represent each side using a Venn diagram, and then check that the diagrams match.





A

$B \cup C$

$A \cap (B \cup C)$

$A \cap B$

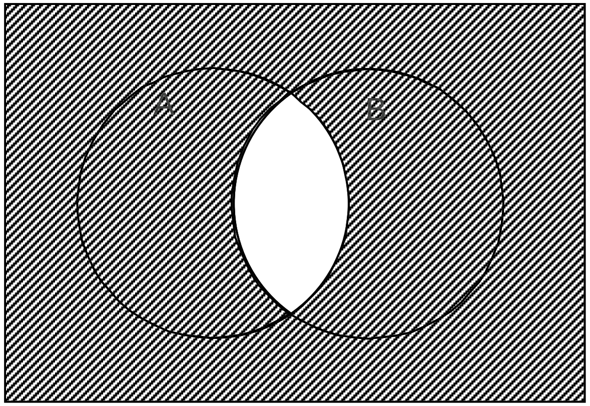
$A \cap C$

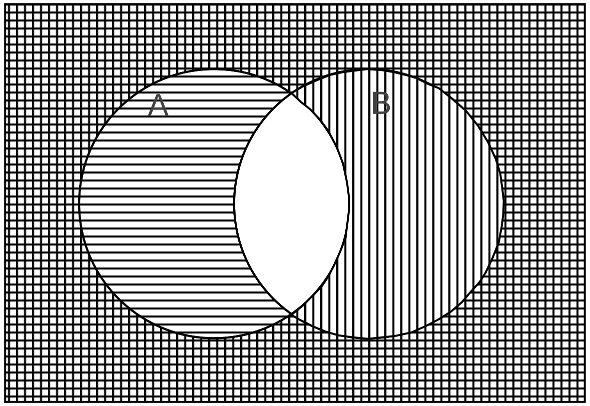
all shaded: $(A \cap B) \cup (A \cap C)$

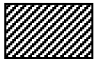
Illustration of laws using Venn diagrams

Another example:

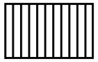
$$(A \cap B)' = A' \cup B'$$



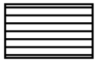




$(A \cap B)'$



A'



B'

all shaded: **$A' \cup B'$**

Analogy with logic concepts

The laws are essentially the same, but are just presented differently.

$\sim(A \vee B) \equiv \sim A \wedge \sim B$ corresponds to $(A \cup B)' = A' \cap B'$

$\sim$ corresponds to $'$

$\wedge$ corresponds to $\cap$

$\vee$ corresponds to $\cup$

T corresponds to **U**

F corresponds to $\varnothing$

Power set of a set

The power set of a set is the set of all subsets of the set.

For example, if $S = \{a, b\}$ then the power set of S is

$$P(S) = \{\{\ \}, \{a\}, \{b\}, \{a, b\}\}$$

When creating a subset of a set, with each element we have the choice of whether to include it or not. So if the set has n elements, then it has 2^n subsets.

Example:

Suppose there are three days of the week I can do yoga: Mon, Wed, Fri.

I could choose to go to some, all or none of the yoga sessions. The possibilities are: none, M, W, F, M&W, M&F, W&F and all.

In terms of sets, the choices are:

{ }, {M}, {W}, {F}, {M, W}, {M, F}, {W, F}, {M, W, F}

These are all the possible subsets of the set {M, W, F}, and therefore they comprise the power set of {M, W, F}.

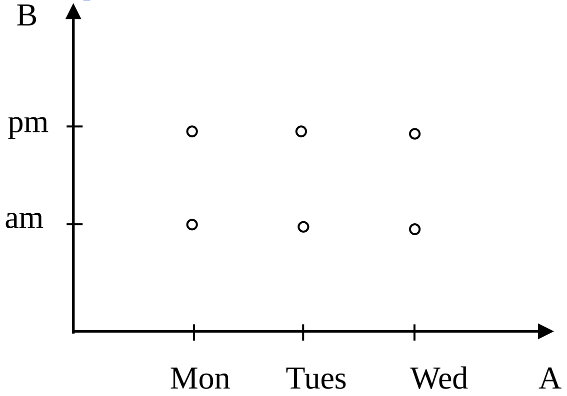
Cartesian Product

When working with graphs we talk about Cartesian coordinates.
Each point is represented by an ordered pair of numbers; that is a pair of numbers where the order is important.
So, for example, the point (2, 5) is different from the point (5,2).

The Cartesian product of sets A and B, denoted by $A \times B$, is the set of ordered pairs where the first member of the pair comes from A and the second from B.

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$$

Example:
Let $A = \{\text{Mon, Tues, Wed}\}$
and $B = \{\text{am, pm}\}$
Then



$A \times B = \{(\text{Mon, am}), (\text{Mon, pm}), (\text{Tues, am}), \ldots, (\text{Wed, pm})\}$

The End

Lecture’s Major Objectives

After completing this section, students should be able to:

- translate from a verbal statement involving **universal and existential quantifiers** into a symbolic statement, and vice versa
- find the negations of quantified statements
- draw **Venn diagrams** to represent **set expressions**
- find the complement of a set
- find the union and intersection of two sets
- find the Cartesian product of two sets and draw its graph
- find the power set of a set
- illustrate the equality of sets using Venn diagrams
- apply the Addition Principle